Coherent versus Measure-First Copying of Phase-State Codebooks: Exact Leak-Order Laws

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Abstract

Phase-state codebooks $\psi_x = d^{-1/2} \sum_j \omega^{xk_j} |j\rangle$ are the states used in quantum fingerprinting and quantum hashing. We compare, for every copying task $T \to T + k$, the optimal coherent copier (any quantum channel) and the optimal measure-first copier (entanglement-breaking: measure, then prepare) on a finite codebook with the same quantities for the continuous family of all phase states. Both finite-codebook optima are at least the continuous ones; if the keys have no additive collisions among multisets of size $2T + k$, both are equal for every dimension; and for codebooks that are collision-free at size T, every collision at size $2T + k$ strictly increases the value of the natural measure-first strategy. For cyclic qubit codebooks of q states and every $1 \to m$ task ($m \geq 2$) we prove an exact leak-order theorem: the measure-first optimum equals the continuous one if and only if $q \geq m + 2$, while the coherent optimum already equals it when $q \geq m + 1$, so $q = m + 1$ separates the two attackers for every m, beginning with three qubit states at $1 \to 2$. For one input copy and two or three output copies we determine the measure-first optimum of the continuous family in every dimension $d \geq 3$—for example $(7d - 8 + \sqrt{25d^2 - 64d + 48})/(2d^3)$ at $1 \to 2$—so that for cyclic codebooks in every dimension the coherent $1 \to 2$ optimum equals the continuous one if and only if the keys form a Sidon set, while the measure-first optimum does if and only if they form a B_3 set. We give an explicit polynomial-phase hash with a guaranteed (sufficient) copy budget and prove a separation for it. Whether measure-first attackers leak no later than coherent ones for every codebook remains open; the stronger statement that the measure-first gap always exceeds the coherent gap is false for general finite phase codebooks.

1 Introduction

Earlier work determined exact copying limits for phase-state codebooks [1], for designs [2] and for stabilizer states [3], including what is lost by measuring first. Here we ask which kind of attacker first notices that a codebook is finite rather than continuous. The answer is asymmetric: a coherent copier is sensitive only to additive structure of order $T + k$ in the cases covered by [1, Thm. 9], while a measure-first copier is sensitive to order $2T + k$. Finite alphabets are known to raise classical benchmark thresholds in quantum-memory tests [6, 7]; our results quantify exactly when this happens for phase codebooks, and show that the coherent optimum can stay unchanged while the classical one moves. At $1 \to 1$ the same threshold ($q \geq 3$) is consistent with the observation of Fuchs and Sasaki that regular great-circle ensembles with at least three states have the continuous measure-and-prepare fidelity [5]. Finite sets and continuous families have been compared before in

the asymptotic cloning regime [12], for single-clone fidelity [13] (finite equatorial sets reproducing a continuous phase-covariant cloning objective also appear in a variational photonic implementation [34]), and for measure-and-prepare fidelity [5, 44]; here both optima are exact at finite copy numbers and the two attackers are compared directly. For the continuous family itself, optimal phase-covariant cloning is known in many cases, beginning with [29]: for qubits with arbitrary input and output numbers and for single-input qutrits [14], for qudits with $M-N$ a multiple of d [15], for single-clone fidelity [16], for the global $1 \to 2$ optimum $(2d-1)/d^2$ in every dimension [17], and for qudit phase states with any numbers of inputs and outputs, together with an attaining channel [28]; the relation between global cloning and estimation is studied asymptotically in [12, 18]. In particular, Yang and Chiribella formulated the measure-and-prepare global fidelity with entangled re-preparation allowed, wrote the finite-size objective for equatorial qubits with the covariant measurement, and solved its large-output limit [18]; the qubit measure-first values of Theorem 11 are finite-size instances of that formulation, and Theorem 9 gives exact finite-copy optima at $1 \to 2$ and $1 \to 3$, with optimality certificates, for every $d \geq 3$; together with Theorem 11 for $d = 2$, these cover all $d \geq 2$. Classical benchmarks for phase-covariant states were studied in [39], optimal estimation of multiple phases in [40], phase-covariant devices in arbitrary dimension in [41], and no-signalling bounds linking cloning and metrology in [42]. Phase-state codebooks built from characters are the quantum fingerprints and quantum hashes of [19–22, 30, 31]; related pseudorandom and design constructions are in [24–26]. Classical (noncontextual) models cannot reach the optimal phase-covariant cloning fidelity [35]; coherent versus measure-and-prepare cloning of quantum channels [36] and coherent versus incoherent inference [37] show related separations, and finite sampling sets are used numerically in SDP cloning frameworks [38].

2 Setting

Let G be a finite abelian group with character group $\widehat{G}$, and let $K = (k_1, \ldots, k_d)$ be distinct elements of G (keys), $d \geq 2$. For $x \in \widehat{G}$ put $\psi_x = d^{-1/2} \sum_{j=1}^{d} \chi_x(k_j)|j\rangle$ and $\rho_x = |\psi_x\rangle\langle\psi_x|$, with x uniform on $\widehat{G}$; for $G = \mathbb{Z}_q$, $\chi_x(k) = \omega^{xk}$ with $\omega = e^{2\pi i/q}$. These are character codebooks; more generally, a finite phase codebook is any finite list of states $d^{-1/2} \sum_j e^{i\phi_j}|j\rangle$ with the uniform distribution on the list. The continuous family ("torus") is $\psi_\theta = d^{-1/2} \sum_j e^{i\theta_j}|j\rangle$ (the phase-random states of [43]) with θ distributed by the normalized Haar measure of $(S^1)^d$. A copier for the task $T \to T+k$ $(T, k \geq 1)$ is a channel (completely positive, trace preserving) from operators on $\mathrm{Sym}^T(\mathbb{C}^d)$ to operators on $\mathrm{Sym}^{T+k}(\mathbb{C}^d)$; it is measure-first if it is entanglement-breaking, i.e. of the form $\rho \mapsto \sum_y \mathrm{tr}(M_y\rho)\sigma_y$ [4]. Its average global fidelity on an ensemble μ is $\mathbb{E}_\mu \mathrm{tr}[\rho^{\otimes(T+k)}\Lambda(\rho^{\otimes T})]$. We write P_{coh} and P_{EB} for the optimal values over all channels and over entanglement-breaking channels, and

$$\Delta_{\mathrm{coh}} = P_{\mathrm{coh}}(K) - P_{\mathrm{coh}}(\mathrm{torus}), \qquad \Delta_{\mathrm{EB}} = P_{\mathrm{EB}}(K) - P_{\mathrm{EB}}(\mathrm{torus}).$$

K is B_h if distinct multisets of h keys have distinct sums in G.

Scope. The torus measure is the Haar measure of the phase group $(S^1)^d$, not the Haar measure on all pure states. Proposition 2 and Lemma 1 hold for every finite phase codebook; Theorems 3 and 11, Lemmas 5 and 8 and Section 5 use the character structure; Section 6 concerns general finite phase codebooks. All results use average global fidelity with the uniform distribution on $\widehat{G}$ (Sections 3–5; codebook states may repeat when characters do not separate them) or on the codebook states (Section 6). Restricting copiers to channels from Sym^T to Sym^{T+k} loses nothing: inputs are supported on Sym^T, and since every target state lies in Sym^{T+k}, replacing an output σ

by $\Pi\sigma\Pi + \mathrm{tr}[(1-\Pi)\sigma]\,\sigma_0$ (Π the symmetric projector, σ_0 any symmetric state) does not lower the fidelity and keeps measure-and-prepare channels measure-and-prepare. Copying fidelities, equal or not, are not by themselves statements about forging a token or passing a verifier: no acceptance-probability reduction is given here, and noise is not addressed (the gaps in our examples range from about 10^{-20} at $q = m+1$, $m = 12$, to above 0.25 for the trine at $2 \to 3$). Worst-case, single-copy or verifier-acceptance figures of merit, non-uniform priors, and attackers between measure-first and fully coherent (for example with limited quantum memory) are not covered, and the gaps we find can be very small (Remark after Theorem 11).

Lemma 1 (Objective). *For each fixed channel with Choi operator J, the average global fidelity equals $\mathrm{tr}\big(J\,X^{(T+k,T)}\big)$ with $X^{(T+k,T)} = \mathbb{E}\,\rho^{\otimes(T+k)} \otimes \bar{\rho}^{\otimes T}$, a mixed moment of total order $2T+k$. The optimal values are suprema of these linear functionals over the respective channel sets.*

3 General results

Proposition 2 (Non-negative gaps). *For every codebook and task, $\Delta_{\mathrm{coh}} \geq 0$ and $\Delta_{\mathrm{EB}} \geq 0$.*

Proof. Let Λ be optimal for the torus and let

$$\tilde{\Lambda}(\cdot) = \int U_\theta^{\otimes(T+k)}\,\Lambda\big(U_\theta^{\dagger\otimes T}\cdot\,U_\theta^{\otimes T}\big)\,U_\theta^{\dagger\otimes(T+k)}\,d\theta$$

be its twirl over the diagonal unitaries U_θ. Since the torus is a single orbit of this group, $\tilde{\Lambda}$ has the same fidelity on every phase state, equal to the torus optimum; in particular on the codebook. Twirling preserves entanglement breaking. □

Theorem 3 (No collisions of order $2T+k$). *If K (keys in G) is B_{2T+k}, then*

$$X^{(T+k,T)}(K) = X^{(T+k,T)}(\mathit{torus}),$$

hence $\Delta_{\mathrm{coh}} = \Delta_{\mathrm{EB}} = 0$, for every d.

Proof. In the normalized occupation basis, $|\psi_x\rangle^{\otimes n} = d^{-n/2}\sum_{|\alpha|=n}\sqrt{\mathrm{mult}(\alpha)}\,\chi_x(K\cdot\alpha)|\alpha\rangle$, where $K\cdot\alpha = \sum_j \alpha_j k_j$. The entry of $X^{(T+k,T)}$ indexed by $((b,a),(b',a'))$ with $|b| = |b'| = T+k$, $|a| = |a'| = T$ is $d^{-(2T+k)}\sqrt{\mathrm{mult}(b)\mathrm{mult}(b')\mathrm{mult}(a)\mathrm{mult}(a')}$ times $\mathbb{E}_x\chi_x\big(K\cdot(b-b'-a+a')\big)$, which by character orthogonality is 1 if $K\cdot(b+a') = K\cdot(b'+a)$ and 0 otherwise. For the torus the same entry carries $\mathbb{E}_\theta e^{i\theta\cdot(b+a'-b'-a)}$, which is 1 iff $b+a' = b'+a$ as occupation vectors. Both $b+a'$ and $b'+a$ are multisets of $2T+k$ keys, so B_{2T+k} makes the two conditions coincide. Equal objectives for every channel give equal optima. □

Theorem 4 (Coherent copying, cyclic keys [1, Thm. 9]). *If $K \subset \mathbb{Z}_q$ is B_{T+k} then $\Delta_{\mathrm{coh}} = 0$ when $d = 2$ (all T), and when $T \in \{1,2,3\}$ (all d).*

Lemma 5 (Reduced coherent dual for any finite abelian group). *Let $r_n(g) = \sum_{|\alpha|=n,\,K\cdot\alpha=g}\mathrm{mult}(\alpha)$ and $S = \mathrm{supp}(r_T)$. Then*

$$P_{\mathrm{coh}}(K) = \min_{z\in(0,\infty)^S}\sum_{g\in S}\frac{r_T(g)}{z_g} \quad \textit{subject to} \quad d^{-(2T+k)}\sum_{g\in S} r_{T+k}(s+g)\,z_g \leq 1 \quad (s\in G),$$

and the minimum is attained.

Proof. Order every operator as $H_{\text{out}} \otimes \overline{H_{\text{in}}}$ with $H_{\text{out}} = \mathrm{Sym}^{T+k}(\mathbb{C}^d)$ and $H_{\text{in}} = \mathrm{Sym}^{T}(\mathbb{C}^d)$, as in Lemma 1. The Choi primal $\max\{\mathrm{tr}(JX) : J \geq 0,\ \mathrm{tr}_{\text{out}}\, J = I_{\text{in}}\}$ has the dual $\min\{\mathrm{tr}\, Y :\ I_{\text{out}} \otimes Y \geq X\}$ with Y on H_{in}; $J = I/\dim \mathrm{Sym}^{T+k}$ is strictly feasible, so strong duality holds. By character orthogonality $X = \sum_{s\in G} |w_s\rangle\langle w_s|$ with $|w_s\rangle = d^{-(2T+k)/2} \sum_{K\cdot b - K\cdot a = s} \sqrt{\mathrm{mult}(a)\mathrm{mult}(b)}\, |b\rangle \otimes |a\rangle$, where b is an output and a an input occupation (the input factor carries the complex-conjugate phase). With $W_x = U_x^{\otimes(T+k)}$ and $V_x = U_x^{\otimes T}$ restricted to the symmetric subspaces, X is invariant under conjugation by $W_x \otimes \overline{V_x}$; hence if $I_{\text{out}} \otimes Y \geq X$ then $I_{\text{out}} \otimes (\overline{V_x}\, Y\, \overline{V_x}^{\dagger}) \geq X$, and the average $\bar{Y} = \mathbb{E}_x \overline{V_x}\, Y\, \overline{V_x}^{\dagger}$ is feasible with the same trace. $\bar{Y}$ commutes with every $\overline{V_x}$, whose characters separate the input charge classes, so $\bar{Y} = \bigoplus_{g\in S} Y_g$. On the sector of charge difference s, X is the single rank-one term $|w_s\rangle\langle w_s|$ and $I_{\text{out}} \otimes \bar{Y}$ is a direct sum of copies of Y_g, one for each output occupation b with $K\cdot b = s + g$; so by the rank-one Schur complement $I_{\text{out}} \otimes \bar{Y} \geq X$ is equivalent to $v_g \in \mathrm{range}(Y_g)$ and the displayed inequality with $z_g = v_g^{\dagger} Y_g^{+} v_g$, $v_g = \sum_{K\cdot a = g} \sqrt{\mathrm{mult}(a)}|a\rangle$. Cauchy–Schwarz gives $\mathrm{tr}\, Y_g \geq r_T(g)/z_g$, with equality for $Y_g = v_g v_g^{\dagger}/z_g$. Each z_g is bounded above (some $r_{T+k}(s+g) > 0$) and the objective diverges as $z_g \to 0$, so the minimum is attained in $(0,\infty)^S$. □

Remark 6. Restricting the variables to $S = \mathrm{supp}(r_T)$ matters: allowing variables outside S turns the minimum into an infimum that need not be attained (for $G = \mathbb{Z}_5$, $K = (0,1)$, $T = k = 1$ the value $3/4$ is then only approached).

Lemma 7 (Measure-first optimum of the torus, $T = 1$)**.** *For a unit seed $e \in \mathbb{C}^d$ let $B_{d,m}(e) = \int |\langle e|\psi_\theta\rangle|^2 |\psi_\theta^{\otimes m}\rangle\langle\psi_\theta^{\otimes m}|\, d\theta$. Then $P_{\mathrm{EB}}(\text{torus}; 1 \to m) = d \max_{\|e\|=1} \lambda_{\max}(B_{d,m}(e))$, and the maximum may be taken over non-negative real seeds.*

Proof. Refine the effects of any measure-and-prepare channel into rank-one terms $t_y |e_y\rangle\langle e_y|$ with pure outputs; each contributes at most $t_y \lambda_{\max}(B(e_y))$ and $\sum_y t_y = d$. Conversely, for any unit e and top eigenvector v of $B(e)$, the effects $(d/d!)\, U_\theta P_\pi |e\rangle\langle e| P_\pi^{\dagger} U_\theta^{\dagger}$ (U_θ diagonal, P_π a coordinate permutation, integrated over θ and summed over π) form a POVM, since phase averaging removes off-diagonal terms and permutation averaging makes each diagonal entry $1/d$; preparing the matching rotation of v attains $d\lambda_{\max}(B(e))$. Writing $d^{m+1} B(e) = D + \sum_{i\neq j} \bar{e}_i e_j A_{ij}$ with D diagonal and A_{ij} entrywise non-negative, the Rayleigh quotient is maximized, entry by entry, by replacing e with $|e|$. □

Lemma 8 (Every collision helps the natural attack)**.** *Let $D_T = \dim \mathrm{Sym}^T(\mathbb{C}^d)$, $n = 2T + k$, $\mathrm{mm}(a)$ the multinomial coefficient of an occupation a, and $\varphi = D_T^{-1/2} \sum_{|a|=T} |a\rangle$ the flat seed. For a character codebook whose keys K (in G) are B_T, the effects $E_x = (D_T/|\widehat{G}|)\, U_x^{\otimes T} |\varphi\rangle\langle\varphi| U_x^{\dagger\otimes T}$, $U_x = \mathrm{diag}(\chi_x(k_j))$, form a POVM on $\mathrm{Sym}^T(\mathbb{C}^d)$; preparing $U_x^{\otimes(T+k)}|v\rangle$ on outcome x gives fidelity $D_T\, \langle v|N_K|v\rangle$ for unit v, where*

$$(N_K)_{b,b'} = \frac{\sqrt{\mathrm{mm}(b)\mathrm{mm}(b')}}{D_T\, d^n} \sum_{|a|=|a'|=T} \sqrt{\mathrm{mm}(a)\mathrm{mm}(a')}\ \big[K\cdot(a+b) = K\cdot(a'+b')\big]$$

$$(|b| = |b'| = T + k).$$

For the torus the same construction (effects $D_T\, U_\theta^{\otimes T} |\varphi\rangle\langle\varphi| U_\theta^{\dagger\otimes T} d\theta$) is a POVM (phase averaging removes every off-diagonal term) and gives N_{tor}, with the indicator replaced by $[a + b = a' + b']$. Then $N_K \geq N_{\text{tor}}$ entrywise, N_{tor} is irreducible, and if K has a collision among multisets of size $2T + k$ then $\lambda_{\max}(N_K) > \lambda_{\max}(N_{\text{tor}})$; hence the natural strategy does strictly better on the codebook.

Proof. Completeness: $\sum_x E_x = D_T \sum_{a,a'} \varphi_a \bar{\varphi}_{a'}\, \mathbb{E}_x \chi_x(K\cdot a - K\cdot a') |a\rangle\langle a'|$; since K is B_T, the average vanishes unless $a = a'$, leaving $D_T \sum_a |\varphi_a|^2 |a\rangle\langle a| = I$. Fidelity: by covariance every

outcome contributes equally, giving $D_T\,\mathbb{E}_y|\langle\varphi|\psi_y^{\otimes T}\rangle|^2|\langle\psi_y^{\otimes(T+k)}|v\rangle|^2 = D_T\langle v|N_K|v\rangle$, and expanding both overlaps in the occupation basis and averaging the characters gives the displayed entries. Entrywise order: $a+b=a'+b'$ implies $K\cdot(a+b)=K\cdot(a'+b')$. A collision $\gamma\neq\gamma'$ of size $2T+k$, written $\gamma=a+b$, $\gamma'=a'+b'$, adds a positive term to $(N_K)_{b,b'}$ that is absent from N_{tor}. $(N_{\mathrm{tor}})_{b,b'}>0$ whenever b' arises from b by moving one unit of occupation ($T\geq 1$), which connects all occupations, so N_{tor} is irreducible with a strictly positive Perron vector u, and $\lambda_{\max}(N_K)\geq u^{\mathsf{T}}N_K u > u^{\mathsf{T}}N_{\mathrm{tor}}u=\lambda_{\max}(N_{\mathrm{tor}})$. □

R0. We call R0 the statement that the flat seed is optimal for the torus, that is,

$$P_{\mathrm{EB}}(\mathrm{torus}) = D_T\,\lambda_{\max}(N_{\mathrm{tor}}).$$

Where R0 holds, Theorem 3 and Lemma 8 give, for B_T codebooks, $\Delta_{\mathrm{EB}}=0 \iff K$ is B_{2T+k}.

Scope	Status of R0
$d=2$, $T=1$, every k	proved (Lemma 7 and the qubit argument of Theorem 11)
$T=1$, $k=1,2$ ($1\to2$, $1\to3$), every $d\geq 3$	proved: Theorem 9 (analytic dual certificates for all d)
$T=1$: $d=3$, $k\leq 8$; $d=4$, $k\leq 7$; $d=5$, $k\leq 5$; $d=6$, $k\leq 4$; $d=7$, $k\leq 3$; $d=8$, $k\leq 2$. Also $d=5$, $2\to3$	proved by exact zero-gap dual certificates (Grok Bot), each checked by an independently written exact checker (ChatGPT's, run by Grok Bot and by Claude)
general d, $T=1$, $k\geq 3$	open. Partial results, with $M(p)=d^{m+1}B_{d,m}(\sqrt{p})$ for a probability vector p (Lemma 7; R0 says $\lambda_{\max}M(p)$ is largest at uniform p): in the auxiliary $1\to1$ case ($k=0$, outside the tasks considered here) the flat seed is optimal in every dimension (ChatGPT; checked by Claude); for every integer $r\geq1$, $\mathrm{tr}\,M(p)^r$ is a symmetric polynomial in p with non-negative coefficients, and in exact computations for small d,m,r it has non-negative coefficients in the elementary symmetric polynomials of p; if that held for all r, R0 would follow, since $\lambda_{\max}M=\lim_r(\mathrm{tr}\,M^{2r})^{1/(2r)}$ (ChatGPT, derived, not yet refereed)

Consequently the equivalence $\Delta_{\mathrm{EB}}=0 \iff B_{m+1}$ for cyclic codebooks at $1\to m$ is proved for $d=2$ (all m), for $m=2,3$ in every dimension, and for $d=3$ ($m\leq 9$), $d=4$ ($m\leq 8$), $d=5$ ($m\leq 6$), $d=6$ ($m\leq 5$), $d=7$ ($m\leq 4$) and $d=8$ ($m\leq 3$).

Theorem 9 (R0 at $1\to2$ and $1\to3$ in every dimension)**.** *For every $d\geq 3$ the measure-first optimum of the continuous family is attained by the flat seed:*

$$P_{\mathrm{EB}}(\mathit{torus};1\to2)=\frac{7d-8+\sqrt{25d^2-64d+48}}{2d^3},\qquad P_{\mathrm{EB}}(\mathit{torus};1\to3)=\frac{\rho(A_d)}{d^4},$$

where $\rho(A_d)$ is the largest eigenvalue of $A_d=\begin{pmatrix}24d-54 & 18 & 0\\ 6(d-2) & 6d-3 & 1\\ 0 & 3(d-1) & d\end{pmatrix}$, the largest real root of $Y^3-(31d-57)Y^2+(174d^2-564d+381)Y-144d^3+576d^2-612d+162$. (At $d=2$ both values are covered by Theorem 11.)

Proof outline. The flat seed is achievable (Lemma 7); its value is the Perron root of the type-reduced flat-seed matrices $\frac{1}{d^3}\begin{pmatrix}6d-8 & 2\\ 2(d-1) & d\end{pmatrix}$ and A_d/d^4. Upper bounds come from explicit dual certificates of the PPT relaxation, given in closed form for every d in Appendix B, whose blocks are shown positive semidefinite for all $d\geq 3$. □

Corollary 10 (Leak-order laws in every dimension)**.** *Let $K \subset \mathbb{Z}_q$ be a cyclic codebook of distinct keys in any dimension d. At $1 \to 2$: $\Delta_{\mathrm{coh}} = 0$ if and only if K is a Sidon set [1, Thm. 7], and $\Delta_{\mathrm{EB}} = 0$ if and only if K is B_3; hence every Sidon set that is not B_3 separates the two attackers. At $1 \to 3$: $\Delta_{\mathrm{coh}} = 0$ if K is B_3 (Theorem 4), and $\Delta_{\mathrm{EB}} = 0$ if and only if K is B_4.*

Proof. The "if" directions are Theorems 3 and 4; the measure-first "only if" directions follow from Lemma 8 (every collision of order $m+1$ strictly raises the flat-seed value of K) and Theorem 9 (the flat seed is optimal for the continuous family; for $d = 2$, Theorem 11). □

4 The leak-order theorem for qubit codebooks

Theorem 11 (Leak order)**.** *Let $F_q = \{(|0\rangle + \omega^j|1\rangle)/\sqrt{2} : j \in \mathbb{Z}_q\}$, $q \geq 2$, and consider $1 \to m$, $m \geq 2$. Then*

$$P_{\mathrm{EB}}(F_q) = P_{\mathrm{EB}}(\textit{torus}) \iff q \geq m+2, \qquad q \geq m+1 \implies P_{\mathrm{coh}}(F_q) = P_{\mathrm{coh}}(\textit{torus}).$$

Hence on this class $\Delta_{\mathrm{coh}} > 0 \Rightarrow \Delta_{\mathrm{EB}} > 0$, and $q = m+1$ gives $\Delta_{\mathrm{EB}} > 0 = \Delta_{\mathrm{coh}}$ for every $m \geq 2$.

Proof. Reduction. The group generated by the phase rotations by $2\pi/q$ and the basis swap acts irreducibly on $\mathbb{C}^2$ (for $q = 2$ it contains the Pauli group), and it maps F_q and the torus to themselves. Refining any measure-first channel into rank-one effects $t_y|\phi_y\rangle\langle\phi_y|$ ($\sum t_y = 2$) and twirling over this group, as in Lemma 7, gives $P_{\mathrm{EB}}(F_q) = 2\max_\phi \lambda_{\max} B_q(\phi)$ with $B_q(\phi) = \mathbb{E}_{\psi \in F_q} |\langle\phi|\psi\rangle|^2 |\psi^{\otimes m}\rangle\langle\psi^{\otimes m}|$, and likewise for the torus.

Seed. For $\phi = (u_0, u_1)$, $|\langle\phi|\psi_\theta\rangle|^2 = \frac{1}{2} + |u_0 u_1|\cos(\theta - \eta)$ with $|u_0 u_1| \leq \frac{1}{2}$. In the Dicke basis the entries of $B_q(\phi)$ are averages of these Fourier modes against $e^{i\theta(a-b)}$, so $|B_q(\phi)_{ab}| \leq B_q(\phi_0)_{ab}$ for the flat seed $\phi_0 = (|0\rangle + |1\rangle)/\sqrt{2}$; $B_q(\phi_0)$ is entrywise non-negative, so $\lambda_{\max} B_q(\phi) \leq \lambda_{\max} B_q(\phi_0)$. This holds for every q and for the torus (this is R0 at $d = 2$).

Matrix. Explicitly $B_q(\phi_0)_{ab} = \frac{\sqrt{\binom{m}{a}\binom{m}{b}}}{2^{m+2}}\big(2[a \equiv b] + [a - b \equiv 1] + [a - b \equiv -1]\big)$, congruences mod q; for the torus, equalities. Since $|a - b| \leq m$, $B_q = B_{\mathrm{tor}}$ iff $q \geq m+2$. For $q \leq m+1$ some entry is strictly larger and B_{tor} is irreducible and tridiagonal, so Perron–Frobenius gives a strict increase.

Coherent half. $\{0, 1\} \subset \mathbb{Z}_q$ is B_m when $q \geq m+1$; apply Theorem 4 with $d = 2$. □

Remark 12. The coherent condition $q \geq m+1$ is sufficient, not necessary: coherent equality can also hold for smaller q (compare [1, Thm. 18]; Grok Bot found exact cases). The separation at $q = m+1$ holds for every m but shrinks quickly: $\Delta_{\mathrm{EB}} \approx 0.0407$ at $m = 2$ and $\approx 1.8 \cdot 10^{-20}$ at $m = 12$ (exact computation by Grok Bot).

Codebook, task	coherent: codebook vs. torus	measure-first, codebook	measure-first, torus
trine F_3, $1 \to 2$	$3/4 = 3/4$	$(7 + \sqrt{17})/16 \approx 0.69519$	$(3 + \sqrt{5})/8 \approx 0.65451$
F_4, $1 \to 3$	equal (Theorem 11)	$(3 + \sqrt{3})/8 \approx 0.59151$	$(11 + \sqrt{61})/32 \approx 0.58782$
trine F_3, $2 \to 3$	$\geq (65 + 42\sqrt{2})/128 > (7 + 4\sqrt{3})/16$	$\geq (65 + 42\sqrt{2})/128 \approx 0.97185$	$0.7131710\ldots$ (Section 6)

For the trine at $2 \to 3$ both gaps are positive: the codebook's measure-first value exceeds the torus's coherent value $(7 + 4\sqrt{3})/16$ (computed exactly by Grok Bot; it also follows from the phase-state formula of [28]), which bounds both torus optima. The lower bound $(65+42\sqrt{2})/128$ comes from the optimal two-copy minimum-error measurement followed by three copies of the guess (ChatGPT); Grok Bot certified the sharper ≥ 0.9718569 for a trine with phases rounded to rationals (within about 10^{-12}), which we do not use.

5 A polynomial-phase hash with a guaranteed copy budget

For a prime p and $N \geq 2$ let $r = 2N - 1 < p$, and take as messages the polynomials $f \in \mathbb{F}_p[t]$ with $\deg f \leq r$ and $f(0) = 0$ (p^r messages), with $|\psi_f\rangle = p^{-1/2} \sum_{t \in \mathbb{F}_p} \omega^{f(t)} |t\rangle$, $\omega = e^{2\pi i/p}$. Equivalently: keys $k_t = (t, t^2, \ldots, t^r)$ in $G = \mathbb{Z}_p^r$ (the moment curve), characters indexed by coefficient vectors. QH-N is thus a character-based quantum hash in the sense of [20, 22], with keys on the moment curve; polynomial phases with Weil-bound overlap control also appear in [23].

Theorem 13 (QH-N). *(a) For $a + b \leq r$: $\mathbb{E}_f \rho_f^{\otimes a} \otimes \bar{\rho}_f^{\otimes b} = \mathbb{E}_\theta \rho_\theta^{\otimes a} \otimes \bar{\rho}_\theta^{\otimes b}$. (b) Every task $T \to T + k$ with $T + k \leq N$ has the torus optima, coherent and measure-first. (The budget N is sufficient, not necessary, and is not a security guarantee by itself.) (c) Distinct messages give distinct rays, and $|\langle\psi_f|\psi_g\rangle| \leq (r-1)/\sqrt{p}$ for $f \neq g$ (informative only when $p > (r-1)^2$). (d) From T copies of a uniformly random message, any measurement identifies it with probability at most* $\min\{1, \binom{p+T-1}{T}/p^r\}$.

Proof. (a) An entry of the moment is a constant times $\mathbb{E}_f \omega^{\sum_t c_t f(t)}$ with c the difference of two multisets X, Y of size $a + b$ in $\mathbb{F}_p$; averaging over the independent uniform coefficients of $t, \ldots, t^r$ gives 1 iff $\sum_t c_t t^j \equiv 0$ for $1 \leq j \leq r$, i.e. X, Y have equal power sums of orders $1..r \geq a + b$. Since $a + b < p$, Newton's identities give equal elementary symmetric functions, so $X = Y$; this is the torus condition. (b) By Lemma 1 the objective uses the moment of order $2T + k \leq 2N - 1 = r$. (c) $\langle\psi_f|\psi_g\rangle = p^{-1} \sum_t \omega^{h(t)}$ with $h = g - f \neq 0$, $h(0) = 0$. Modulus 1 would force all $\omega^{h(t)}$ equal to $\omega^{h(0)} = 1$, so h would have p roots, impossible for $1 \leq \deg h < p$. For $\deg h = 1$ the sum vanishes; for $2 \leq \deg h \leq r < p$ the Weil bound [11] gives $|\sum_t \omega^{h(t)}| \leq (\deg h - 1)\sqrt{p}$. (d) For a POVM $\{E_f\}$ on $\mathrm{Sym}^T(\mathbb{C}^p)$, $\sum_f \mathrm{tr}(E_f \rho_f^{\otimes T}) \leq \sum_f \mathrm{tr}\, E_f = \dim \mathrm{Sym}^T(\mathbb{C}^p)$. □

Part (a) is an instance of exact phase (toric) designs [8–10], close to the bounded-independence phase states of [24, 26]; (c) uses the Weil bound as in [23]; (d) is a standard counting bound, not tight for $T \geq 2$ (sharper bounds follow from [10, Thm. 4.3]). What we add is the reading of (b) as a copy budget for both attackers and the separation below; (b)–(d) do not by themselves give a forgery bound. The Welch bound is $\mu_W = \sqrt{(M-d)/(d(M-1))}$ for M states in dimension d [27]. Observed worst-case overlaps are 1.8–2.0 times the Welch bound for $r = 3$ and up to 3.51 times for $r = 5$ ($7 \leq p \leq 61$; Grok Bot; Claude reproduced $r = 3$, $11 \leq p \leq 31$, and $r = 5$, $p = 11, 13$).

Proposition 14 (QH-N separation). *For $p = 5$, $r = 3$ (budget $N = 2$) and the task $2 \to 3$ (beyond the budget): $P_{\mathrm{coh}}(QH\text{-}N) = P_{\mathrm{coh}}(torus) = (205 + 16\sqrt{39})/625$, while $P_{\mathrm{EB}}(QH\text{-}N) \geq 0.2156168 > P_{\mathrm{EB}}(torus) = \lambda$, where $\lambda \approx 0.1998044540314200455$ is the largest real root of*

$$93132257461547851 5625\, x^6 - 34749507904052734 3750\, x^5 + 40472126007080078125\, x^4$$
$$- 1804318725585937500\, x^3 + 33468360117187500\, x^2 - 234659289600000\, x + 542664896544,$$

isolated in the interval

$$\left[\tfrac{892350922891786}{4466121274511029},\ \tfrac{136625708187137}{683797109776402}\right].$$

Hence $\Delta_{\mathrm{EB}} \geq 0.0158 > 0 = \Delta_{\mathrm{coh}}$.

Evidence. Coherent equality: Lemma 5 applied to $G = \mathbb{Z}_5^3$ with the torus-optimal dual point, which remains feasible (exact). Torus measure-first value: an exact zero-gap dual certificate of the PPT relaxation, matching the flat-seed lower bound. QH-N lower bound: an explicit flat-seed POVM with exact rational preparation vector. Each step was checked by a second system (Appendix A). □

For reference (values computed by Grok Bot; $1 \to 2$ and $1 \to 3$ at $d = 5, 7$ and $2 \to 3$ at $d = 5$ reproduced by Claude by SDP; $(667 + 36\sqrt{38})/2401$ reproduced by Claude through the reduced dual of Lemma 5), the torus coherent optima are $9/25$ (the global $1 \to 2$ optimum $(2d-1)/d^2$ of [17], also [1]), $24/125$ and $(205 + 16\sqrt{39})/625$ for $1 \to 2$, $1 \to 3$, $2 \to 3$ at $d = 5$, and $13/49$, $36/343$, $(667 + 36\sqrt{38})/2401$ at $d = 7$ (all six follow from the optimal cloning fidelity of phase states given, with an attaining channel, by [28]; we computed them independently. Their optimum is for worst-case fidelity; it equals the average-fidelity optimum used here because the optimal channel can be taken covariant, as in Proposition 2); the QH-N measure-first values at $1 \to 2$ (within budget, equal to the torus) are $(27 + \sqrt{353})/250$ ($p = 5$) and $(41 + 5\sqrt{33})/686$ ($p = 7$) (Grok Bot; reproduced by Claude).

6 The H-SEP conjecture and a refuted strengthening

Conjecture 15 (H-SEP). *For every finite phase codebook and every task,* $\Delta_{\mathrm{coh}} > 0 \Rightarrow \Delta_{\mathrm{EB}} > 0$*: measure-first attackers notice a finite alphabet no later than coherent ones.*

It holds on the class of Theorem 11. Evidence (producer-reported search, Grok Bot): 576 search records at $d = 2$ for $2 \to 3$, $2 \to 4$, $3 \to 4$ and at $d = 3$ for $2 \to 3$, plus one trine validation certificate (577 certificate files). For 565 of them the bundled exact checker certifies $\Delta_{\mathrm{EB}} > 0$ from the stored witnesses (re-run by Claude and Grok Bot); the remaining 12 are rational approximations of regular q-gons (Grok Bot), for which Theorem 3 predicts zero gap for the exact q-gons; their certificates neither confirm nor exclude a gap below about 10^{-8}. Of the 576 records, 549 are non-group families and 27 are rational approximations of regular q-gons (15 with a certified positive gap, 12 uncertified). The generating search code is not part of the bundle; the search is exploratory evidence, not an exhaustive or independently reproduced search.

The strengthening $\Delta_{\mathrm{EB}} \geq \Delta_{\mathrm{coh}}$ is false. All certified violations below are for general finite phase codebooks of qubits at $2 \to 3$. For a six-state qubit codebook, $\Delta_{\mathrm{coh}} \geq 0.0898415$ and $\Delta_{\mathrm{EB}} \leq 0.0775753$, so $\Delta_{\mathrm{EB}}/\Delta_{\mathrm{coh}} \leq 0.8634685$; the certificates give the sharper 0.8634671 (Grok Bot's certificates; Claude's SDPs reproduce the coherent values and the PPT upper bounds on the measure-first values, numerically). The torus measure-first value these certificates use lies in $[P_\flat, U]$, where $P_\flat = \frac{1}{4} + \frac{3\sqrt{2}}{32} + \frac{\sqrt{61+36\sqrt{2}}}{32} \approx 0.7131710241$ is the exact value of the flat-seed strategy of Lemma 8 (Claude) and U is Grok Bot's certified upper bound, a window of width $2 \cdot 10^{-11}$. Grok Bot also certified an eight-state codebook with ratio in $[0.8451338, 0.8451344]$ and a fourteen-state codebook with ratio at most 0.8561526 (for the eight-state certificate ChatGPT re-ran Grok Bot's coherent checker together with its own exact measure-first lower bound; both cases were reproduced numerically by Claude). H-SEP holds in all three. Any universal constant c with $\Delta_{\mathrm{EB}} \geq c\,\Delta_{\mathrm{coh}}$ therefore satisfies $c \leq 0.8451344$. At $d = 3$ the smallest ratio found was 1.1342847 (Grok Bot; reproduced numerically by Claude; partial search). For fixed d, T, k and number of states, if H-SEP holds then a positive constant exists on any region with Δ_{coh} bounded below (a compactness argument by ChatGPT, not yet refereed).

7 Open problems

(1) R0 for every d at $T = 1$ and $m \geq 4$ (now known for $m = 2, 3$ and all d), which would extend Corollary 10 to every m; (2) H-SEP for $T \geq 2$ and for codebooks that are not B_T; (3) the optimal constant c, or a sequence of codebooks with ratio tending to 0; (4) a security reduction turning these

copying bounds into forgery bounds for a copy-budgeted token (compare quantum signatures with limited key copies [32] and optimal counterfeiting of Wiesner money [33]); (5) the qubit theorem for $T \geq 2$; (6) the same comparison for other figures of merit and for attackers between measure-first and fully coherent.

A Provenance and reproducibility

Result	Derived by	Checked by (independent unless marked as a re-run)
Prop. 2	ChatGPT	Claude
Thm. 3, Lemmas 7, 8	ChatGPT	Grok Bot (line-by-line referee; exact cases)
Lemma 5 (any finite abelian group)	ChatGPT	Grok Bot (referee; 6 exact instances)
Thm. 11	ChatGPT	Grok Bot (referee; exact for $m \leq 12$); Claude (SDP values)
R0 certificates ($d = 3$–8)	Grok Bot (a first $d = 4$, $k = 5$ dual was infeasible and was replaced)	ChatGPT (own exact checker); Claude ran it on all 30 certificates (logs in bundle)
Prop. 14	Grok Bot (exact values, torus certificates); ChatGPT (coherent-equality dual argument, exact rational lower bound)	each by the other; Claude (SDPs; re-ran both checkers)
Thm. 13	Claude (construction; ChatGPT proposed the polynomial family)	ChatGPT (referee); Grok Bot (checks)
Thm. 9 (all d)	ChatGPT	Grok Bot (line-by-line referee, exact checks $d \leq 40/20$, independent fit); ChatGPT fresh session (hostile referee); Claude (PPT numerics)
Search, H-SEP+ counterexamples	Grok Bot	Claude (re-ran all 577 search certificates, log in bundle; SDP reproduction of the six-, eight-, fourteen-state and $d = 3$ values, numerical); ChatGPT (re-run of Grok Bot's coherent checker with its own exact measure-first lower bound for the six-state, eight-state and a further six-state violation)

"Re-ran" means executing another system's checker; "independent" means separately written code or a line-by-line referee. Grok Bot also measured the Welch ratios, pointed to [10], showed that the copy budget is sufficient but not necessary, certified the bound 0.9718569 for a rationalised trine at $2 \to 3$ (not used here), and corrected an earlier "first" to "no later than". Certificates, independent checkers and expected logs are collected in one archive,

`hsep_qhn_certificates.zip`,

whose SHA-256 is

`2d9d96c45902b036bd1495569a0d972f5199567dc15fa22eb01611c633708413`

Its `README.md` gives one command and the expected output for each item; `MANIFEST_SHA256.txt` lists every file. On release, the archive is deposited with this preprint's Zenodo record and in the folder `paper4/` of `https://github.com/Weldog/sidon-phase-cloning` (tag `paper4-v1`); any copy can be checked against the hash above.

B Dual certificates for Theorem 9

Setting. Output basis: normalized occupation states $|\beta\rangle$ of $\mathrm{Sym}^m(\mathbb{C}^d)$, $h_\beta = m!/\prod_j \beta_j!$. The fidelity operator is $R_{(i,\beta),(j,\beta')} = \sqrt{h_\beta h_{\beta'}}\, d^{-(m+1)}\,[\beta - e_i = \beta' - e_j]$. For $W \succeq 0$ and $Y = y\,I$, if $Z = Y \otimes I - R - W^\Gamma \succeq 0$ then every PPT-feasible Choi matrix, in particular every measure-first channel, has fidelity at most $\mathrm{tr}\,Y = dy$. After conjugating by $\mathrm{diag}(h_\beta^{-1/2})$, R has the constant entry $C = d^{-(m+1)}$ inside each difference-charge block and W is constant on S_d-orbits w_j inside each sum-charge block. A single feasible pair (W, Z) with dy equal to the flat-seed value proves optimality; no uniqueness of the dual is needed.

$1 \to 2$. Let $X = \frac{1}{2}(7d - 8 + \sqrt{25d^2 - 64d + 48})$, the Perron root of $\left(\begin{smallmatrix} 6d-8 & 2 \\ 2(d-1) & d \end{smallmatrix}\right)$, so that $X^2 - (7d-8)X + 6d^2 - 12d + 4 = 0$, equivalently $(X-d)(X-6d+8) = 4(d-1)$. Take $y = X/d^4$ and

$$w_0 = 0, \quad w_1 = \tfrac{(d-1)(X-6d+8)}{4d^4}, \quad w_2 = -\tfrac{d-1}{d^4}, \quad w_3 = \tfrac{X-d}{d^4}, \quad w_4 = \tfrac{X-2d}{2d^4}, \quad w_5 = \tfrac{2d-X}{4d^4}.$$

The sum-charge blocks are $W_{[3]} = [w_0] = 0$, $W_{[21]} = \left(\begin{smallmatrix} w_3 & w_2 \\ w_2 & w_1 \end{smallmatrix}\right)$ (determinant 0 by the identity; diagonal positive) and $W_{[111]}$ with diagonal w_4 and off-diagonal w_5 (eigenvalues 0, $\frac{3(X-2d)}{4d^4}$ twice). The difference-charge singletons of Z vanish; the $d \times d$ block of charge [1] has eigenvalue $\eta = \frac{(d-2)(6d-X)+8}{4d^4}$ on a $(d-2)$-dimensional transverse space and the remaining 2×2 sector $\frac{1}{d^4}\left(\begin{smallmatrix} (X-6d+8)/4 & -\sqrt{d-1} \\ -\sqrt{d-1} & X-d \end{smallmatrix}\right)$, of determinant 0. For $d \geq 3$, $2d < X < 6d$ (from the closed form), so $\eta > 0$ and $X - 2d > 0$; and $X - 6d + 8 = 4(d-1)/(X-d) > 0$ by the identity, so all diagonals are positive. Hence $W, Z \succeq 0$ and $dy = X/d^3$, the flat-seed value.

$1 \to 3$. Let $Y = \rho(A_d)$, $r = \frac{Y-d}{3(d-1)}$, $s = \frac{(Y-6d+3)r-1}{6(d-2)}$ (so $(s, r, 1)$ is the positive Perron vector of A_d), $C = d^{-4}$, $y = Y/d^5$, $L = \frac{d-1}{d^5(Y-d)}$. The kernel equations leave one free parameter; we take the solution on which the two orbit values w_3, w_4 both vanish, and set

$$\begin{aligned} &w_0 = 0,\ w_1 = \tfrac{(d-1)^2}{d^5(Y-d)},\ w_2 = -\tfrac{d-1}{d^5},\ w_3 = w_4 = 0,\ w_5 = \tfrac{Y-d}{d^5}, \\ &w_8 = \tfrac{Y-3d}{3d^5},\ w_{10} = \tfrac{Y-6d}{6d^5},\ w_{11} = -\tfrac{w_{10}}{3}, \\ &w_9 = \tfrac{w_8 - L}{d-2} - C, \quad w_6 = \tfrac{r^2}{2s^2}(w_8 + w_9), \quad w_7 = -\tfrac{s}{r}\,w_6 \end{aligned}$$

(w_{10}, w_{11} absent at $d = 3$). Every kernel equation reduces to the Perron equation for A_d. Positivity: $Y > 9d$ for all $d \geq 3$ (at $d = 3$ the leading 2×2 block already has Perron root 27; for $d \geq 4$, $Y > 24d - 54 \geq 9d$); hence $r, s > 0$, $w_8 > 2C$, $w_1 < C/8$, $0 < L < C/(8(d-1))$; moreover $s/r > \sqrt{3}$ at $d = 3$ and $s/r > 2$ for $d \geq 4$, which gives $0 < w_6 < w_8/3$, $w_8 \pm w_9 > 0$ and $y/3 + w_9 > 0$. Every block is then zero, rank one PSD, or has the explicit nonnegative eigenvalues listed in the bundled proof, and $dy = Y/d^4$ equals the flat-seed value.

Verification. Derived by ChatGPT; refereed line by line by Grok Bot (first-principles rebuild of every block; exact checks $d = 3$–40 at $1 \to 2$ and $d = 3$–20 at $1 \to 3$, with corrupted controls rejected) and by ChatGPT in a fresh session as a hostile referee (first-principles rebuild of the physical dual operators, $11 + 12$ symbolic identities); Grok Bot fitted the same $1 \to 2$ closed forms independently; Claude's direct numerical solution of the PPT relaxation matches both values at $d = 3$–7 ($1 \to 2$) and $d = 3$–5 ($1 \to 3$). All scripts and logs are in `R0_all_d/` of the certificate archive.

Use of AI tools

This work was carried out with Grok Bot (SpaceXAI), ChatGPT (OpenAI), Claude (Anthropic) and Perplexity and Consensus (literature searches), as detailed in Appendix A. The author directed the work and takes full responsibility for its content.

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